3 JavaScript and TypeScript deep dive

This chapter covers

- JavaScript fundamentals
- TypeScript features and trends

Having reviewed the core web technologies of HTML and CSS, Jane turned her focus to JavaScript and TypeScript—two pillars of modern web development that play a critical role in technical interviews.

JavaScript, a high-level, versatile language, has evolved far beyond its origins as a simple browser scripting tool. Today, it powers dynamic, feature-rich applications across the web, server, and even native platforms. Standardized by ECMA International and guided by the TC39 committee, JavaScript continues to grow with new proposals and features. See the latest specification at https://tc39.es/ecma262 and learn more about the TC39 process in Appendix A.2.

TypeScript (https://www.typescriptlang.org) has risen rapidly as a must-have skill for frontend developers. Its static type system catches errors early, improves code quality, and empowers developers with intelligent tooling like autocompletion and type-aware refactoring. As more organizations embrace TypeScript for UI development, proficiency in it is now a common expectation in technical interviews, reflecting its central role in building scalable applications.

Let’s review some JavaScript and TypeScript skills. For Jane, this was not her first time preparing for JavaScript and TypeScript interviews. She already knew the major features, but each focused review revealed new patterns, pitfalls, and advanced insights that sharpened her understanding and deepened her expertise.